

JOSHUA CARBON

COMPUTER ENGINEER / QA AUTOMATION

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SUMMARY

Computer Engineering graduate with practical experience in QA automation, software development, embedded systems, and IoT. Built a web automation framework using Python, Robot Framework, Selenium, and SeleniumLibrary with reusable components, automated execution, reporting, and optional AI-powered test analysis. Comfortable with debugging, test design, and technical documentation.

SKILLS

Programming Languages: Python, C#, C++, C

QA & Automation: Robot Framework, Selenium, SeleniumLibrary, Manual Testing, Test Case Design, Debugging

Software Development: Object-Oriented Programming, Windows Forms, Software Testing, Quality Assurance (QA)

Web Development: React, Typescript, Vite, React Router, HTML, CSS

Tools: Git, GitHub, Postman, Microsoft Visual Studio, Visual Studio Code

AI / Test Analysis: OpenAI API, LLM-assisted Test Analysis

Embedded Systems & IoT: Arduino, Raspberry Pi, ESP32, Blynk, Sensor Integration, Embedded C

PROJECTS

QA Automation Framework — Built a structured web automation framework using Python, Robot Framework, Selenium, and SeleniumLibrary, featuring reusable automation components, automated test execution, reporting, and optional AI-powered analysis of test results.

Inventory Management System — Developed a desktop-based inventory management application using C# and Windows Forms for managing product records and inventory workflows.

MotorPH Payroll Management System — Developed a desktop-based payroll management system using C# and object-oriented programming for employee records, attendance, payroll computation, leave management, and reporting.

AI-Based Japanese Character Recognition using YOLOv5 — Developed a Raspberry Pi-based computer vision system using a custom-trained YOLOv5 model to recognize Japanese Kanji characters in real time.

Personal Portfolio Website — Built and deployed a portfolio site using React, TypeScript, and Vite, including a resume selector and project case-study pages — self-tested across devices and browsers before launch.

EXPERIENCE

Intern/OJT — Erovoutika Robotics and Automation Solution | May 2023 – Aug 2023

- Collaborated with engineering teams to identify and troubleshoot technical issues across automation systems, contributing to improved system reliability.
- Performed debugging, calibration, and functional testing on automation processes, helping catch and resolve issues before deployment.
- Assisted in integration testing of sensors and embedded components, verifying functionality across multiple automation initiatives.
- Documented test procedures and technical guidelines, improving consistency and knowledge transfer within the team.

EDUCATION

Bachelor of Science, Computer Engineering — Mapúa University | 2019 – 2025

Bachelor of Science, Electrical Engineering | 2014 – 2019

ADDITIONAL

- Familiar with Software Development Life Cycle (SDLC) and willing to learn.
- Familiar with Object-Oriented Programming (OOP) concepts and eager to deepen understanding.
- Interested in Artificial Intelligence, Machine Learning, Automation, and related engineering technologies.